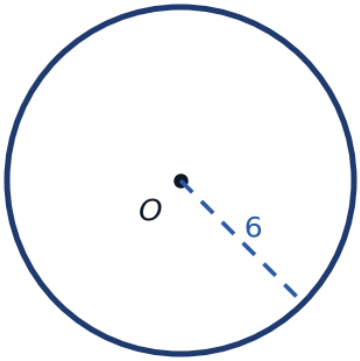
Circles & Coordinate Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

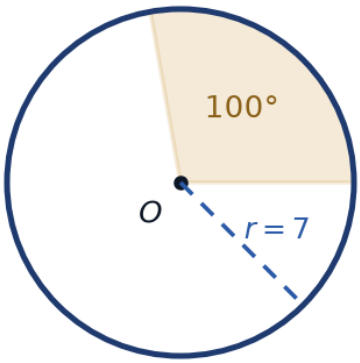
1 [Foundational] • Target: The equation of a circle

The circle below has the radius shown. Find its area and circumference, in terms of π.



2 [Foundational] • Target: Circle theorems

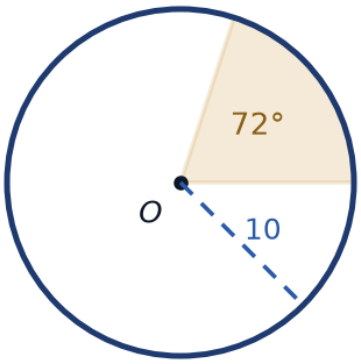
A chord subtends the central angle shown below. Find the measure of the inscribed angle subtending the same (minor) arc.



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3 [Foundational] • Target: Arc length and sector area

The circle below has the radius and shaded sector shown. Find the area of the sector, in terms of π.



4 [Foundational] • Target: The equation of a circle

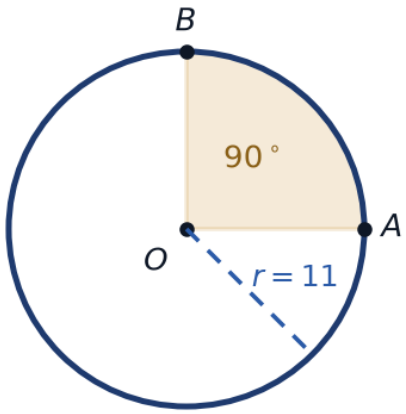
Write the equation of a circle with center $(-4, 7)$ and radius 9.

5 [Foundational] • Target: The equation of a circle

A circle has diameter 26 and is centered at the origin. Write its equation, and state its radius.

6 [Foundational] • Target: Completing the square

Find the center and radius of $x^2 + y^2 - 6x + 8y - 11 = 0$.



7 [Foundational] • Target: Completing the square

Find the center and radius of $x^2 + y^2 + 10x - 4y + 13 = 0$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Distance, midpoint, and slope**

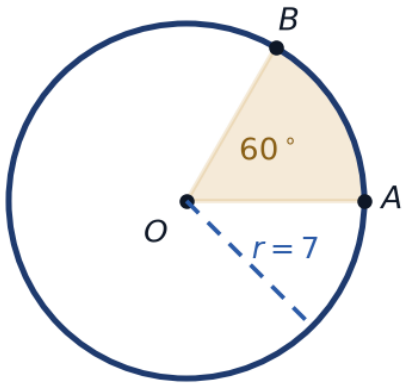
Find the distance between $(-3, 5)$ and $(4, -7)$, to the nearest hundredth.

9 **[Intermediate]** • **Target: Distance, midpoint, and slope**

Find the midpoint and slope of the segment joining $(2, -6)$ and $(-8, 10)$.

10 **[Intermediate]** • **Target: Distance, midpoint, and slope**

Find the slope of the line through $(-2, 7)$ and $(6, -5)$, and state whether the line rises or falls from left to right.



11 **[Intermediate]** • **Target: Coordinate proofs**

Show that the triangle with vertices $A(0, 0)$, $B(4, 0)$, $C(4, 3)$ is a right triangle, using slopes.

12 **[Intermediate]** • **Target: Coordinate proofs**

Show that $A(1, 2)$, $B(4, 3)$, $C(3, 6)$, $D(0, 5)$ form a parallelogram, by showing both pairs of opposite sides are parallel.

13 **[Intermediate]** • **Target: Tangent lines to circles**

A circle has center $O(2, 3)$ and radius 5. Point $P(2, 12)$ lies outside the circle. Find the length of the tangent from P to the circle, to the nearest hundredth.

14 **[Intermediate]** • **Target: Tangent lines to circles**

Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$ on the circle.

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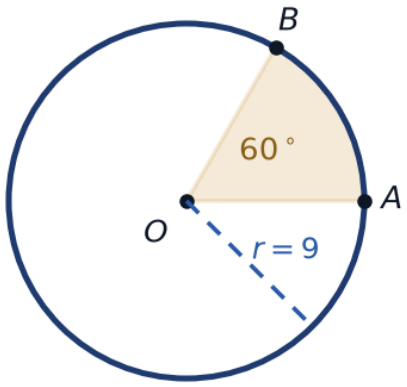
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Tangent lines to circles • Cross-Lesson: 7.01 Solid Geometry & Similarity

A circle has center $(0, 0)$ and radius 10. Determine whether the point $(6, 9)$ lies inside, on, or outside the circle.

16 [Challenging] • Target: Circle theorems • Cross-Lesson: 8.01 Statistics & Probability

A central angle of 84° subtends an arc. Find the measure of the inscribed angle subtending the same arc.



17 [Challenging] • Target: Arc length and sector area • Cross-Lesson: 1.01 Advanced Algebra

A circle has radius 12 cm. Find the arc length subtended by a central angle of $\frac{5\pi}{6}$ radians.

18 [Challenging] • Target: Arc length and sector area • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

A sector of a circle with radius 9 has area 27π. Find the central angle, in radians.

19 [Challenging] • Target: Conics: ellipses and hyperbolas • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Identify the type of conic and state its center for $\frac{(x-2)^2}{16} + \frac{(y+1)^2}{9} = 1$.

20 [Challenging] • Target: Conics: ellipses and hyperbolas • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Identify the type of conic, its center, and its asymptote slopes for $\frac{(x+3)^2}{4} - \frac{y^2}{9} = 1$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The equation of a circle

Write the equation of a circle with center $(9, 12)$ and radius 12.

a. $(x-9)^2+(y-12)^2=12$

b. $(x+9)^2+(y+12)^2=144$

c. $(x-9)^2+(y+12)^2=144$

d. $(x-9)^2+(y-12)^2=144$

22 [Foundational] • Target: Completing the square

Write the equation of a circle with center $(8, -7)$ and radius 7.

a. $(x-8)^2+(y-7)^2=49$

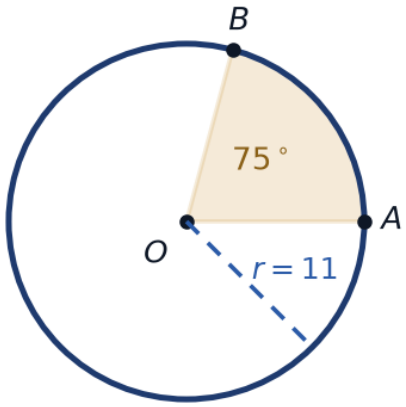
b. $(x+8)^2+(y-7)^2=49$

c. $(x-8)^2+(y+7)^2=7$

d. $(x-8)^2+(y+7)^2=49$

23 [Foundational] • Target: Distance, midpoint, and slope

Write the equation of a circle with center $(11, -9)$ and radius 5.



a. $(x+11)^2+(y-9)^2=25$

b. $(x-11)^2+(y+9)^2=25$

c. $(x-11)^2+(y+9)^2=5$

d. $(x-11)^2+(y-9)^2=25$

24 [Foundational] • Target: Coordinate proofs

Write the equation of a circle with center $(0, 10)$ and radius 10.

a. $(x)^2+(y-10)^2=102$

b. $(x)^2+(y-10)^2=100$

c. $(x)^2+(y-10)^2=10$

d. $(x)^2+(y+10)^2=100$

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25 [Foundational] • Target: Tangent lines to circles

Write the equation of a circle with center $(2, 5)$ and radius 12.

a. $(x-2)^2+(y-5)^2=12$

b. $(x-2)^2+(y+5)^2=144$

c. $(x+2)^2+(y+5)^2=144$

d. $(x-2)^2+(y-5)^2=144$

26 [Foundational] • Target: Circle theorems

Write the equation of a circle with center $(-1, 9)$ and radius 11.

a. $(x+1)^2+(y-9)^2=121$

b. $(x-1)^2+(y+9)^2=121$

c. $(x+1)^2+(y+9)^2=121$

d. $(x+1)^2+(y-9)^2=11$

27 [Foundational] • Target: Arc length and sector area

Write the equation of a circle with center $(11, -5)$ and radius 13.

a. $(x+11)^2+(y-5)^2=169$

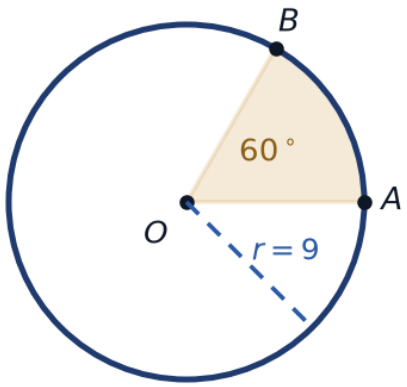
b. $(x-11)^2+(y+5)^2=13$

c. $(x-11)^2+(y-5)^2=169$

d. $(x-11)^2+(y+5)^2=169$

28 [Foundational] • Target: Conics: ellipses and hyperbolas

Find the center and radius of $x^2+y^2+2x-2y-62=0$.



a. $(-1, -1), 8$

b. $(-1, 1), 8$

c. $(1, 1), 8$

d. $(-1, 1), 10$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: A well-designed word problem: a Ferris wheel**

Find the center and radius of $x^2 + y^2 - 6x - 2y - 90 = 0$.

a. $(3, 1), 12$

b. $(-3, 1), 10$

c. $(3, 1), 10$

d. $(3, -1), 10$

30 **[Intermediate]** • **Target: General Circle Equations & Completing the Square**

Find the center and radius of $x^2 + y^2 - 8x - 18y + 81 = 0$.

a. $(-4, 9), 4$

b. $(4, -9), 4$

c. $(4, 9), 6$

d. $(4, 9), 4$

31 **[Intermediate]** • **Target: Tangent Lines to Circles**

Find the center and radius of $x^2 + y^2 - 10x + 8y + 37 = 0$.

a. $(5, 4), 2$

b. $(-5, -4), 2$

c. $(5, -4), 2$

d. $(5, -4), 4$

32 **[Intermediate]** • **Target: Loci & Conic Definitions**

Find the center and radius of $x^2 + y^2 + 4x + 14y - 91 = 0$.

a. $(-2, -7), 12$

b. $(2, -7), 12$

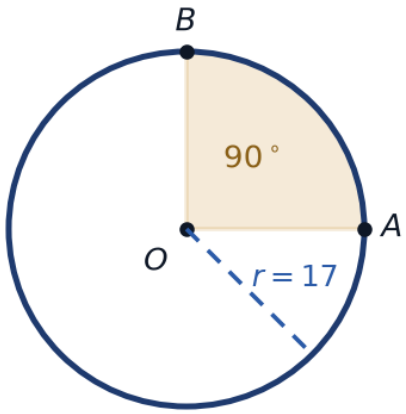
c. $(-2, -7), 14$

d. $(-2, 7), 12$

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33 [Intermediate] • Target: Coordinate Proofs & Polygons

Find the center and radius of $x^2 + y^2 + 12x - 18y + 17 = 0$.



a. $(-6, 9), 12$

b. $(6, 9), 10$

c. $(-6, 9), 10$

d. $(-6, -9), 10$

34 [Intermediate] • Target: The equation of a circle

Find the center and radius of $x^2 + y^2 - 2x + 4y - 116 = 0$.

a. $(1, -2), 13$

b. $(-1, -2), 11$

c. $(1, 2), 11$

d. $(1, -2), 11$

35 [Intermediate] • Target: Completing the square

A circle has center $O(-6, 7)$ and radius 5. Point $P(-6, 20)$ lies outside the circle. Find the length of the tangent from P to the circle.

a. 13

b. 14

c. 12

d. 10

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36 [Intermediate] • Target: Distance, midpoint, and slope

A circle has center $O(7, 5)$ and radius 5. Point $P(7, 18)$ lies outside the circle. Find the length of the tangent from P to the circle.

a. 13

b. 12

c. 10

d. 14

37 [Intermediate] • Target: Coordinate proofs

A circle has radius 13. Find the area of the sector with central angle 40°, in terms of π.

a. $\frac{338}{9}\pi$

b. $\frac{178}{9}\pi$

c. $\frac{160}{9}\pi$

d. $\frac{169}{9}\pi$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Tangent lines to circles • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

A circle has radius 12. Find the area of the sector with central angle 150°, in terms of π.

a. 60π

b. 61π

c. 120π

d. 59π

39 [Challenging] • Target: Circle theorems • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Find the distance between $(0, -7)$ and $(-1, 1)$, to the nearest hundredth.

a. 9.06

b. 7.06

c. 8.06

d. 10.06

40 [Challenging] • Target: Arc length and sector area • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Find the distance between $(5, 9)$ and $(-2, 1)$, to the nearest hundredth.

a. 9.63

b. 12.63

c. 10.63

d. 11.63

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41 [Challenging] • Target: General Circle Equations & Completing the Square • Cross-Lesson: 5.01 Trigonometry

A circle has the equation $x^2 + y^2 + 4x - 10y + 13 = 0$. What are its center and radius?

a. Center $(-2, 5)$, radius 4

b. Center $(2, -5)$, radius 4

c. Center $(-2, 5)$, radius 16

d. Center $(2, -5)$, radius 16

42 [Challenging] • Target: General Circle Equations & Completing the Square • Cross-Lesson: 7.01 Solid Geometry & Similarity

For what value of k does $x^2 + y^2 - 6x + 8y + k = 0$ represent a single point (degenerate circle)?

a. 25

b. 16

c. 9

d. 0

43 [Challenging] • Target: Tangent Lines to Circles • Cross-Lesson: 8.01 Statistics & Probability

At what points does the line $y = x + 2$ intersect the circle $x^2 + y^2 = 10$?

a. $(1, 3)$ and $(-3, -1)$

b. $(3, 1)$ and $(-1, -3)$

c. $(1, 3)$ only

d. No intersection

44 [Challenging] • Target: Loci & Conic Definitions • Cross-Lesson: 1.01 Advanced Algebra

What is the locus of all points whose distances to $(2, 0)$ and $(-2, 0)$ have a constant sum of 6?

a. Ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$

b. Hyperbola $\frac{x^2}{9} - \frac{y^2}{5} = 1$

c. Circle $x^2 + y^2 = 9$

d. Parabola $y^2 = 4x$

45 [Challenging] • Target: Coordinate Proofs & Polygons • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

What type of quadrilateral has vertices at $A(0, 0), B(4, 0), C(5, 3),$ and $D(1, 3)$?

a. Parallelogram

b. Rhombus

c. Rectangle

d. Trapezoid (not parallelogram)

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